PROTIK BISWAS

AI/ML RESEARCH | MEDICAL AI | SOFTWARE ENGINEERING & QA

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PROFESSIONAL SUMMARY

Computer Science researcher with completed BSc coursework at AIUB (**CGPA: 3.94/4.00**). Research spans efficient medical-image segmentation, synchronized biosignal learning, and endoscopy vision-language model reliability. Combines hands-on PyTorch research with software development, test design, and QA automation.

EDUCATION

BSc in Computer Science | American International University-Bangladesh (AIUB)
Coursework completed November 2026; degree conferral pending | CGPA: 3.94/4.00
Higher Secondary and Secondary School Certificates | Milestone School and College | Science | GPA: 5.00/5.00

TECHNICAL SKILLS

Machine learning: Python, PyTorch, TensorFlow, scikit-learn, timm, Hugging Face Transformers, OpenCV, CatBoost, pandas, NumPy, SciPy.
Medical AI: Image classification and segmentation, multimodal learning, transfer learning, Grad-CAM, attention models, calibration and evaluation.
Software and QA: Selenium WebDriver, Pytest, JUnit, test design, functional and regression testing, mutation testing, Git.
Development and data: C++, Java, C#, JavaScript, PHP, SQL, HTML/CSS, MySQL, Oracle, SQL Server, WordPress, Figma, SPSS.

EXPERIENCE

Software Intern | Trends Bird Limited

- Translated requirements into responsive page layouts and high-fidelity Figma prototypes.
- Incorporated design feedback and prepared implementation handoff materials.

SELECTED RESEARCH AND THESIS WORK

Colorectal Polyp Segmentation | INCISCOS V 2026, Paper 65
Boundary-guided EfficientNet-B0 model with size-aware sampling on Kvasir-SEG. The accepted manuscript reports **0.9213 Dice**, **0.8731 IoU**, **4.804M parameters**, and **43.7 FPS** on the held-out test split.

DeepLabNet | Liver Ultrasound Segmentation
Haar-frequency fusion, entropy-guided refinement, and boundary supervision for liver ultrasound; **0.8641 Dice** and **0.7707 IoU**. Accepted at IECBES 2026.

Classifier-Grounded Re-prompting (CGR) | Undergraduate Thesis
Studies whether calibrated image-classifier evidence can guide a second vision-language model response to reduce fabricated or omitted findings in endoscopy descriptions. Evaluation uses HyperKvasir and Kvasir-VQA.

Improved Deep PACFNet | Synchronized ECG-PCG Classification
Lightweight dual-branch deep learning for multimodal cardiac anomaly detection using synchronized ECG and PCG signals; accepted at IEEE SPICSCON 2026.

SOFTWARE ENGINEERING AND QA PROJECTS

- **Hostel Management System QA:** designed 32 test cases, automated key workflows with Selenium and Pytest, and validated PHP/MySQL records.
- **Budget-Aware Mutation-Guided LLM Testing:** compared baseline, repair, random, and adaptive feedback for JUnit generation with Qwen2.5-Coder, Defects4J, and Bugs.jar under fixed resource budgets.
- **Hostel Management System:** built registration, room management, booking, payment, and administration workflows using PHP, MySQL, HTML, CSS, and JavdaScript.

RESEARCH, PUBLICATIONS & ENGINEERING

Selected projects and conference contributions

ADDITIONAL RESEARCH PROJECTS

CerebriLite-CAM | Brain Tumor MRI Classification
MobileNetV2 with CBAM attention and Grad-CAM interpretation for four-class MRI classification; reported **97.0% accuracy**. Presented at COMPAS 2026, Paper 634.

Rare Disease Onset Prediction | Orphadata
Combined genetic and epidemiological variables with TF-IDF text features and Node2Vec disease-gene graph representations using CatBoost. Accepted at the 4th International Conference on Computing Advancements.

LiteRCF-Net | Thyroid Ultrasound Classification
Combined lesion, contextual, and frequency branches; reported **86.71% accuracy** and **0.9065 AUROC** on TN5000.

RESEARCH PUBLICATIONS AND CONFERENCE CONTRIBUTIONS

- **Lightweight Boundary-Guided Colorectal Polyp Segmentation with Size-Aware Training.** Accepted for presentation, INCISCOS V 2026, Paper 65; Quito, Ecuador, November 17-19, 2026.
- **DeepLabNet: Lightweight Liver Ultrasound Segmentation via Haar Frequency Fusion and Entropy Guided Boundary Refinement.** Accepted for presentation, IECBES 2026.
- **Improved Deep PACFNet: Multimodal Deep Learning for Cardiac Anomaly Detection Using Synchronized ECG-PCG Signals.** Accepted, IEEE SPICSCON 2026.
- **CerebriLite-CAM: A Lightweight Interpretable Attention Network for Brain Tumor Classification from MRI.** Presented, COMPAS 2026, Paper 634.
- **Predicting Rare Disease Onset Categories Using Integrated Genetic, Epidemiological, and Graph-Based Features from Orphadata.** Accepted, 4th International Conference on Computing Advancements.
- **Vaccination Coverage and Challenges in Achieving Universal Immunization in Bangladesh.** Contribution recognized at the 7th International Conference on Integrated Sciences, 2025.

ADDITIONAL DEVELOPMENT AND DESIGN

- Interactive WordPress theme and UI/UX prototypes in Figma; Java application (TechGeek); C# supermarket-management system; microprocessor-based heart-rate detector; front-end and C++ projects.

AWARDS, CERTIFICATIONS AND LANGUAGES

- **Awards:** Dean's List; merit-based bachelor's scholarship.
- **Certification and membership:** Cisco Networking certification; Engineering Students' Association of Bangladesh member.
- **Languages:** Bangla (native), English (fluent), Hindi (conversational).

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